

Warm up:

1. Let $X \sim B\left(20, \frac{1}{3}\right)$. Find:
 - a) $P(X = 0)$
 - b) $P(X = 5)$
 - c) $P(X = 18)$
 - d) $P(X > 2)$
2. I roll what I know to be a fair die twenty times. What is the probability that I get:
 - a) Only non-multiples of three
 - b) Five multiples of three
 - c) All but two multiples of three
 - d) More than two multiples of three
 - e) Comment on the above.

Calculator Section:

3. Sterling decides to enter an archery competition. The probability that he hits a bulls-eye is 0.45 and may be assumed to be independent from one shot to the next.
 - a) Determine the probability that, in a series of shots at 15 targets, he hits:
 - i) at most 5 targets;
 - ii) more than 10 targets;
 - iii) exactly 6 targets;
 - iv) at least 5 but at most 10 targets.
 - b) How many targets would you expect him to hit?
 - c) In fact, Sterling hits five targets when he fires. Calculate the probability of this happening and comment of what you notice.
 - d) How many targets would Sterling have to hit to make you doubt the given probability? Explain your answer with reference to specific probabilities.
4. A company manufactures low cost lightbulbs which they sell in boxes of fifty. They know that some of them will be faulty but that the probability of any given lightbulb being faulty is 0.02. What is the minimum number of light bulbs that they can say will work such that they are right at least 90% of the time?
5. A pharmaceutical company is producing vaccines but they know the process they are doing will only work 60% of the time. They need 7 vaccines and how many times should they do the process in order to be 98% sure they will get the vaccines they need?
6. I throw four dice and flip nine coins. I get a point for every six I roll and for every head I throw. What is the probability that I score exactly 4 points?